Charotar University of Science and Technology
M.Sc. (Nano Science & Technology) Semester I~~V~~Examination May 2011

NT809 Nanotherapeutics

Date 2nd May 2011 **Time: 10.00 am to 10.30 pm** **Max. Marks 20**

Part – A

Choose the Right Answer

1. A quantum Dot is:
   a. a semiconductor whose excitons are confined in all three spatial dimensions
   b. The sharpest possible tip of an Atomic Force Microscope
   c. Unexplained spots that appear in electron microscopy images of nanostructures smaller than 1 nanometer
   d. A traditional organic dye
2. Artificially prepared vesicles made of lipid bilayers are called:
   a. Micelles
   b. Reverse Micelles
   c. Liposomes
   d. ISCOMs
3. Polyplexes are:
   a. Polymeric micelles
   b. Complex of DNA and polymers
   c. Complex of viral antigens and lipids
   d. Lipid complex of DNA
4. Spontaneous emulsification depends on:
   a. Critical Micellar concentration of surfactants
   b. Specific gravity of surfactants
   c. Oil:Water ratio
   d. Concentration of co surfactant
5. A solution of solid lipids and water miscible solvent mixture in water will give:
   a. Liposome
   b. Lipoplexes
   c. Solid lipid nanoparticles
   d. Nanocapsules
6. Of the following which cannot be said to have magnetic properties?
   a. Nickel
   b. Aluminium
   c. Greigite
   d. None of the above

7. An antibody which acts as an opsonin:
    a. IgA
    b. IgM
    c. IgE
    d. IgD
8. The main mechanism of transport of nanoparticles across intestinal epithelium is:
    a. Transcellular
    b. Paracellular
    c. Endocytic Pathway
    d. All of the above
9. The main limiting membrane of the skin for transport of external particles/drug is:
    a. Stratum lucidum
    b. Subcutis
    c. Stratum Corneum
    d. Dermal papilla
10. A homogenous matrix system were the drug is dispersed throughout a particle:
    a. Nanocapsules
    b. Nanosphere
    c. Dendrimers
    d. Magnetic nanoparticles

<u>Fill in the Blanks</u>

1. The Process of attaching Polyethylene glycol to a drug is called ----------------------.
2. Spheroid nanostructures built through repeating chemical synthesis are called -------------.
3. The Barrier to be overcome for drugs targeting the CNS is ------------------
4. Ethanol incorporated with liposome is called --------------------.
5. ------------- is a structure made of cholesterol, Quil A & phospholipids
6. ------------------- are a form of nanosuspensions
7. A type of biodegradable bond ------------------.
8. The delivery system with absence of an aqueous phase capable of forming emulsion is called ----- --------------
9. ---------------------- are used as fluorescent probes for biomolecular and cellular imaging
10. ------------- is a type of natural polymer used for nanoparticle preparation.

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Charotar University of Science and Technology

M.Sc. (Nano Science & Technology) Semester IV Examination May 2011

NT809 Nanotherapeutics

Date: 2^{nd} May 2011 **Time: 10.30 am to 13.00 pm** **Max. Marks 50**

Part B

(Write each section on separate answer book)

Section I

Que.1	(a) Write a note on the different Methods of preparation of Nanotherapeutics.	(6)
	(b) What are the factors affecting the carrier properties of nanoparticles?	(4)
Que.2	(a) Write a note on Microemulsions & Nanoemulsions	(4)
	(b) The Application of nanotherapeutics in cancer therapy with examples	(6).
	OR	
Que.2	(a) What is the role of role of biological barriers? Briefly explain their roles	(4)
	(b) How has nanotechnology helped in overcoming the barriers? Discuss with example.	(6)

Section II

Que.3	(a) Write a note on the types of nanoparticles used in dermal drug delivery. Give an example of nanotechnology application with any skin disorder.	(6)
	(b) List out the challenges of dermal drug delivery.	(4)
Que.4	(a) List and define the different types of nanotherapeutics.	(4)
	(b) Explain the role of any two nanotherapeutics in immunological applications.	(6)
	OR	
Que.4	(a) Define: PEgylation. What are the Common polymers used for the preparation of polymeric micelles and other nanoparticles?	(6)
	(b) Explain the role of nanoparticles in bacterial diseases.	(4)
Que.5	(a) Write a note on the use and application of magnetic nanoparticles.	(6)
	(b) Briefly explain the role of different carbon nanomaterials & Quantum dots.	(4)

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